Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Development of an adaptive welding fixture for reliable laser welding of thin metal sheets*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (), Next-gen Nuclear Power (), Advanced Bio (), Aerospace/Marine (), Hydrogen (√), Cybersecurity (), AI (), Next-gen Communication (), Advanced Robotics & Manufacturing (), Quantum ()			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials		*Electrical & Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Optical Energy–Based Processing Machines*	*Optical Application Devices*	*Laser Processing Machines*
Concept of Proposed Technology	*- Mechanical development and characterization of a welding fixture for leak tight, continuous laser welding of geometrical complex thin metal sheets* *- Analysis, evaluation, selection and integration of measurement technology to monitor the clamping and positioning of the sheets, e.g. force sensors, displacement sensors* *- Development and integration of a control loop to compensate for deviations in the clamping conditions, e.g. gap formation.* *- Validation of the system technology for components using different material thicknesses, e.g. ≤100 µm and ≥300 µm.* *- Investigation of a reduction of subsequent quality control processes, such as leakage testing, microscopy and metallurgical analysis by increased process stability*			
Rationale for Joint R&D Project	*The project focuses on the development and validation of a welding position for the reliable laser welding of thin sheets.. Cooperation is essential for successful development and validation. For the components under consideration — namely, metallic bipolar plates for fuel cells (FC) and electrolyzers (EL) — there are different concepts for FC and EL stacks on the German and Asian markets, particularly in Korea. These differences affect key properties such as component size, material and thickness, geometry, the process chain, and operational and production standards. The collaboration aims to ensure the technical and regulatory applicability of the developed welding and clamping technology, thereby opening up new markets and products.* *The Joint R&D project is also essential from a technical perspective for implementing and validating the content of the*			

	research project. Successful implementation requires expertise in laser welding processes, product manufacturing and market requirements, mechanical design (storage, assembly and guidance of components), sensor technology (for detecting inhomogeneities in pressure and gap formation in thin sheets) and control technology (for compensating for deviations in clamping conditions). Such expertise can only be fully represented by a research consortium.
Key Words	*1. Continuous laser welding for thin metal sheets using an actor-based clamping process.* *2. Detection and compensation of clamping deviations for reduction of scrap rate* *3. Reliable welding process enables reduction of downstream quality control* *4. Validation of the fixture for two different material thicknesses, e.g. ≤100 µm and ≥300 µm.*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	*An adaptive welding device for thin sheets can reduce testing processes and scrap by increasing tolerance to deviations in the production process. At the same time, reducing manual steps shortens throughput time. Current welding devices are often based on experience and frequently require several iterations to ensure functionality, particularly for complex thin sheet metal components such as metallic bipolar plates. They are also inflexible in the event of process deviations and do not feature internal quality control. This also requires the welding device to be redesigned when changes are made to different geometries.*			
R&D Target Specifications	*Organizational:* *1. Developed fixture for laser welding is ready to transfer to Korean partner for production of sheet metal components* *Technical* *1.Single-step welding of complex components, eliminating the need for manual intervention* *2. Reduction of the overall process cycle time by 60 %* *3. Reduction of scrap by 50% through compensation for deviations* *4. Reduction of the need for downstream quality control*			

R&D Plan

Work package	1st Year				2nd Year				3rd Year				Final Year	
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
1. Definition of requirements considering product, production scenario & economics														
2. Concept development for the development of the welding fixture														
2.1 Concept development for actor elements														
2.2 Pre trials for actor elements														
2.3 Concept development for mechanical fixture design														
3. Development of mechanical structure for pressing thin sheets for a joining process, particularly laser welding.														
3.1 Selection, development and integration of suitable actuator elements for clamping														
3.2 Development of structural design & mechanical build of the welding fixture														
3.3 Integration of measurement technology to detect deviations in force distribution & gap formation														
3.4 Integration of measurement technology to geometrical misalignment														
4. Adaptive Process Control & Validation														

	4.1 Integration of active elements to compensate for detected deviations in force distribution														
	4.2 Integration of active elements to compensate for geometrical misalignment														
	4.3 Commissioning of the prototypic device and development/parameterization of the control loop to compensate for deviations.														
	4.4 Validation of the technology at an industrially relevant scale.														

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*To implement the project within the consortium, an equal and cooperative distribution is planned. A precise division of work packages and roles should be made based on the specific competencies within the consortium.* *Fraunhofer IPT has extensive experience of the entire sheet metal processing chain, including forming, laser processing, toolmaking and quality control. This includes mechanical development, control engineering, commissioning and process development. In particular, the Fraunhofer Institute contributes its expertise to the mechanical development of the device, the laser welding process, and commissioning and validation.* *Fraunhofer IPT also has the necessary machine infrastructure for development.* *In addition to press systems and tools for sheet metal forming, the project can make use of laser system technology for welding, including e.g. 100W - 1kW fiber lasers, fixed optics or galvo scanner units with working areas of up to 600x600 mm.*
Expected R&R of Korean SME Partner	*To implement the project within the consortium, an equal and cooperative distribution is planned. A precise division of work packages and roles should be made based on the specific competencies within the consortium. There are opportunities in developing and selecting suitable measurement technology for recording and compensation, and in integrating control technology. Additionally, the various market access requirements and product specifications in European and Asian markets, particularly Korea, ensure the technology's universal application.* *Specific roles and activities of the partners* *- Provision of component geometry or production scenario* *- Pre-development at Fraunhofer IPT, scaling at Korean partner* *- Implementation in the production line*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project aims to advance the development of a welding device for thin sheets. Technically, increased process stability reduces wastage. Adaptive process control also reduces development time when product changes are made. Coordinating technical and commercial requirements continuously with the Korean partner ensures successful integration into the production line.* *From a commercial perspective, improvements in the reproducibility of weld seams reduce rework and scrap, particularly by minimizing the need for downstream quality control. Additionally, the application of the developed technology to other products, such as in battery or electronics manufacturing, is to be examined.* *From an ecological perspective, reduced scrap also leads to a smaller carbon footprint due to lower material requirements. Furthermore, this technology can make an important contribution to the further expansion of the hydrogen infrastructure, particularly PEM-FC and PEM-EC.*

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization